

EMC/EMI Pre-Certification Assessment

EEZ BB3+ STM32H7 CM5 module

<https://github.com/eez-open/eez-bb3plus-cm5-mcu>

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Prepared by

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Executive Summary

This report documents EMC pre-certification radiated-emission measurements performed on the EEZ BB3+ STM32H7 CM5 module unit. Measurements were carried out in an in-house DIY shielded chamber at Intergalaktik d.o.o. and calibrated against a reference device (ULX4M-LS) for which absolute emission values are known from an accredited test report (SiQ Ljubljana T251-0973/25). The frequency range covered is 30 MHz – 1 GHz against EN 55032 Class B limits at 3 m reference distance.

Pre-certification testing identifies likely problem frequencies and mitigation needs before submitting the device to an accredited laboratory. It does not replace formal compliance testing.

Scope of Assessment

This pre-certification assessment covers radiated electromagnetic emissions in the 30 MHz – 1 GHz range and conducted emissions on the AC mains port in the 150 kHz – 30 MHz range (LISN). Wireless module compliance (RED 2014/53/EU) and immunity testing are out of scope of this measurement campaign.

Standards Addressed

Standard	Description
EN 55032:2015+A1:2020	Electromagnetic compatibility of multimedia equipment - Emission requirements
CISPR 32:2015+A1:2019	Electromagnetic compatibility of multimedia equipment - Emission (international)
ANSI C63.4 (reference)	Methods of measurement for radio-noise emissions

Methodology

Direct absolute-level radiated emission measurement requires an accredited test chamber (semi-anechoic, 3 m or 10 m measurement distance) with calibrated antennas and traceable instrumentation. Such facilities are expensive and typically available only in commercial test laboratories.

In place of absolute measurement, this assessment uses the SUBSTITUTION METHOD against a known reference device. The procedure is:

- **Reference device characterization.** An accredited test laboratory (SiQ Ljubljana, report T251-0973/25) measured a ULX4M-LS development board in their certified chamber, producing absolute emission curves over 30 MHz – 1 GHz.
- **In-house chamber scan.** We scan the same ULX4M-LS in our DIY shielded chamber using a TinySA Ultra spectrum analyzer + Intergalaktik EMC PCB antenna at 80 cm. The difference between the SiQ values and our raw scan defines a per-frequency chamber transfer function.
- **Empty chamber baseline.** An empty-chamber scan captures ambient noise and residual coupling so we can distinguish EUT emissions from background.
- **EUT scan.** The device under test is scanned in the same chamber configuration and the chamber transfer function is applied to obtain its estimated far-field emission profile in dB μ V/m at 3 m.
- **Pass/fail evaluation.** The calibrated EUT trace is compared against EN 55032 Class B limits. Frequencies where the EUT exceeds the limit are flagged as exceedances.

Important note on values: the levels reported in this document are RELATIVE to the certified ULX4M-LS reference device, not absolute laboratory measurements. They are best-effort engineering estimates suitable for pre-certification assessment. The substitution method assumes that geometric and antenna-coupling differences between the two scans (ULX4M-LS and EUT) are negligible because both are measured in the same chamber configuration.

Test Setup

Equipment used in this measurement campaign:

Item	Specification
Spectrum analyzer	TinySA Ultra (100 kHz - 6 GHz, USB serial)
Vector network analyzer	LiteVNA 64 (50 kHz - 6.3 GHz, USB serial)
Antenna	Intergalaktik EMC PCB (broadband 30 MHz - 1 GHz, KiCad design); OpenSCAD source at github.com/intergalaktik/EMC-experiments
LNA	EMCgear RFA-30 (+40 dB nominal gain, 30 MHz - 1 GHz)
Shielded enclosure	Custom DIY box (300 x 200 x 200 mm) lined with TDK IFM16-030GB300X200 absorber, aluminium-foil layer, and Mission Darkness TitanRF Faraday Fabric tent (400 x 300 x 300 mm, six-sided full-wrap)
LISN (line impedance stabilisation network)	Intergalaktik-built EEZ LISN r1B2 — two-line CISPR 16-1-2 compliant, 50 Ω / 50 μ H per line, 150 kHz – 30 MHz, 10 A continuous at 250 VAC, 2 $\times$ 50 Ω BNC RF take-off, Schürter PF0030 fused inlet, TVS + GDT transient protection.

Chamber Construction

The shielded enclosure is a layered DIY structure designed for low-cost in-house EMC pre-certification measurements:

- **Inner box.** Custom 3D-printed/laser-cut shell, 300 x 200 x 200 mm (width x height x depth), 2.5 mm wall thickness. Antenna mounting hole centred on the top face, OpenSCAD design open-source on GitHub ([intergalaktik/EMC-experiments](https://github.com/intergalaktik/EMC-experiments)).
- **Absorber layer.** TDK IFM16-030GB300X200 ferrite absorber sheets line the inside of the box, providing damping of internal resonances and reducing standing-wave artefacts.
- **Conductive layer.** Aluminium foil applied externally over the absorber-lined box, forming a continuous Faraday-cage layer.
- **Faraday-fabric tent.** Mission Darkness TitanRF Faraday Fabric tent (400 x 300 x 300 mm) wraps the entire box, including the underside, for additional shielding effectiveness against ambient RF.

Geometry

The receiving antenna (Intergalaktik EMC PCB, 40 mm probe length) is fixed at the top centre of the inner box, projecting downward through the top wall. The EUT is placed inside the box on a non-conductive support such that the antenna-to-EUT separation is approximately 10 - 15 cm depending on EUT physical height.

This geometry is REACTIVE near-field at frequencies below ~ 300 MHz ($\lambda/2\pi$ greater than 16 cm) and far-field above ~ 500 MHz. The substitution method (see Methodology) absorbs this geometric non-uniformity into the chamber transfer function calibrated against the SiQ-certified ULX4M-LS reference, so the relative comparison between EUT and reference remains valid across the full band.

Both H- and V-polarisations are scanned; the maximum value is recorded as the per-frequency emission level.

Measurement Software

All measurements, calibration, and report generation in this assessment are coordinated by the EMC Pre-Cert Measurement App (version 0.17.2-w34.7), an in-house web application developed by Intergalaktik. The application runs on a Raspberry Pi 3 single-board computer that is physically attached to the test bench; the operator interacts with it from any workstation on the same network via a standard web browser.

The application orchestrates the full measurement chain over USB serial: TinySA Ultra spectrum analyzer (sweep configuration, peak-hold acquisition, CSV export), LiteVNA-64 vector network analyzer (binary protocol with software 1-port OSL calibration), and an external EMCgear RFA-30 LNA. Calibration files (antenna factor, LNA gain, cable loss, chamber transfer from SiQ ULX4M-LS reference) are stored per project and applied automatically when traces are visualised or exported.

Calibration Data Status

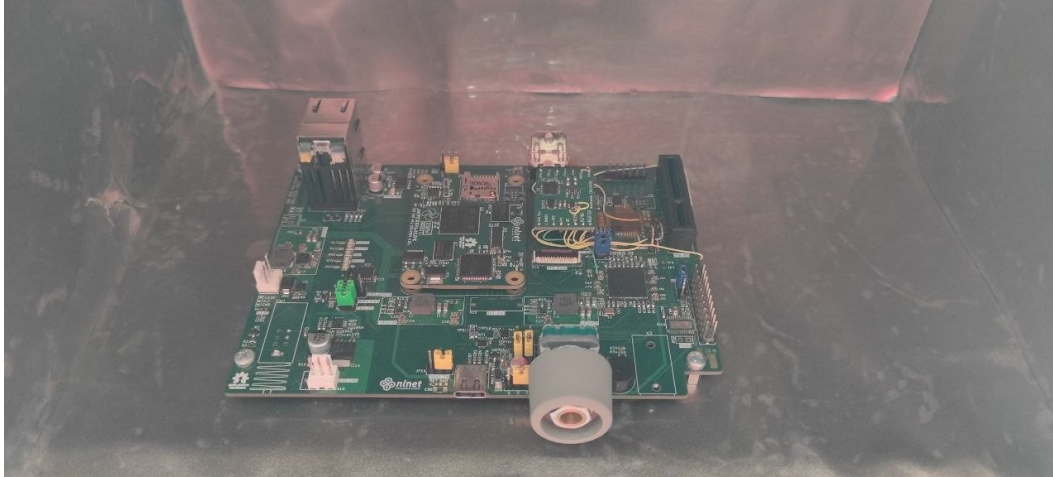
Curve	Source	Points	Span / value used
Antenna factor	measured (project EEZ BB3+ STM32H7 CM5 module)	50	10.0 - 1500.0 MHz
LNA gain	measured (project EEZ BB3+ STM32H7 CM5 module)	401	10.0 - 1500.0 MHz
Cable loss	measured (project EEZ BB3+ STM32H7 CM5 module)	401	10.0 - 1500.0 MHz
Reference device transfer	measured (project EEZ BB3+ STM32H7 CM5 module)	1434	30.0 - 1000.0 MHz

Standard Procedure

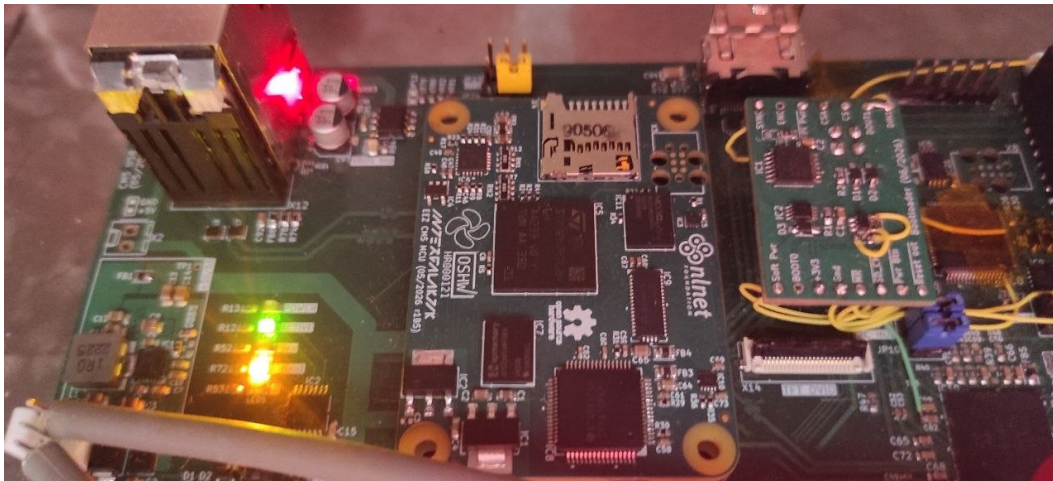
- **1. Project setup.** Create or load a project in the dashboard. Each project owns its own scan folder, calibration data, photos, comments, and reports - making the workflow reproducible across multiple devices under test.
- **2. Calibration.** Upload the SiQ T251-0973/25 reference values to the project's calibration folder. The application derives the chamber transfer function from the in-chamber ULX4M-LS scan, providing per-frequency calibration anchors across 30 MHz - 1 GHz.
- **3. Reference scan (empty chamber).** With the chamber sealed and the EUT removed, run the Reference scan from Quick Actions. The result is the ambient + chamber-residual baseline used to subtract background contributions.
- **4. Reference Device scan.** Place the ULX4M-LS development board in the chamber, supply power, and run the Reference Device scan. This populates the SiQ-anchored transfer function and validates that the chamber matches its calibration profile.
- **5. EUT scan.** Replace the ULX4M-LS with the actual device under test, retain the same antenna geometry, and run the EUT scan. The application stitches 12 frequency segments (each 100 MHz wide, ~3 kHz RBW, 5x peak-hold into a single 30 MHz - 1 GHz CSV).
- **6. Combined Calibrated View.** The dashboard renders all three calibrated traces (Reference, ULX4M-LS, EUT) on a common axis with the EN 55032 Class B limit overlaid, allowing instant visual pass/fail evaluation. A Class B-offset slider lets the operator apply safety margins; the MAX-hold smoothing window pulls narrow peaks above the noise floor for inspection.
- **7. Report generation.** DOCX and PDF reports are generated on demand from the dashboard's Generate buttons. PDFs are produced by LibreOffice headless conversion of the DOCX so both formats are byte-identical in content and layout.

Reproducibility: every scan is a timestamped CSV file under the project folder. Re-running Generate Report at any later time produces a report that reflects the current state of pinned panels, calibration files, photos, and comments, without manual re-editing.

Test Setup Photos



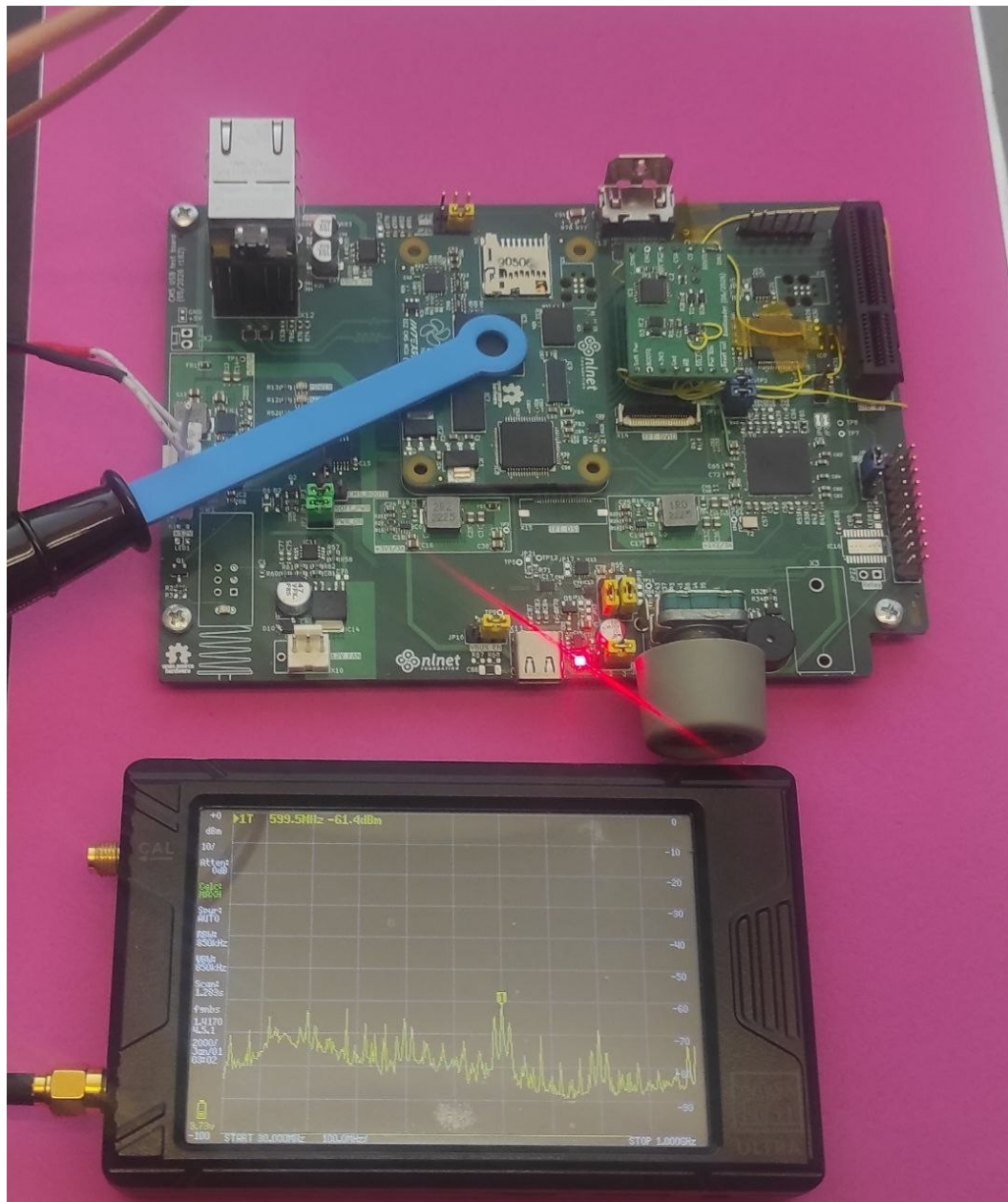
EUT positioned in chamber



Close-up of the EUT (PCB, connectors)



Antenna position in chamber



Near-field probe scanning over EUT



Chamber cover

Combined Calibrated View

The chart below superimposes three calibrated traces (Reference, ULX4M-LS, EUT) against the EN 55032 Class B limit line at 3 m. All three measured traces have been corrected by the chamber transfer function derived from the SiQ-certified ULX4M-LS report and are therefore directly comparable to the limit line.

The ULX4M-LS curve serves as a known-compliant baseline: an EUT that emits at or below the ULX4M-LS curve at every frequency can be expected to also pass the same accredited test, since the ULX4M-LS device already has a passing report. EUT exceedances above the ULX4M-LS curve identify potential compliance risks that need design attention.

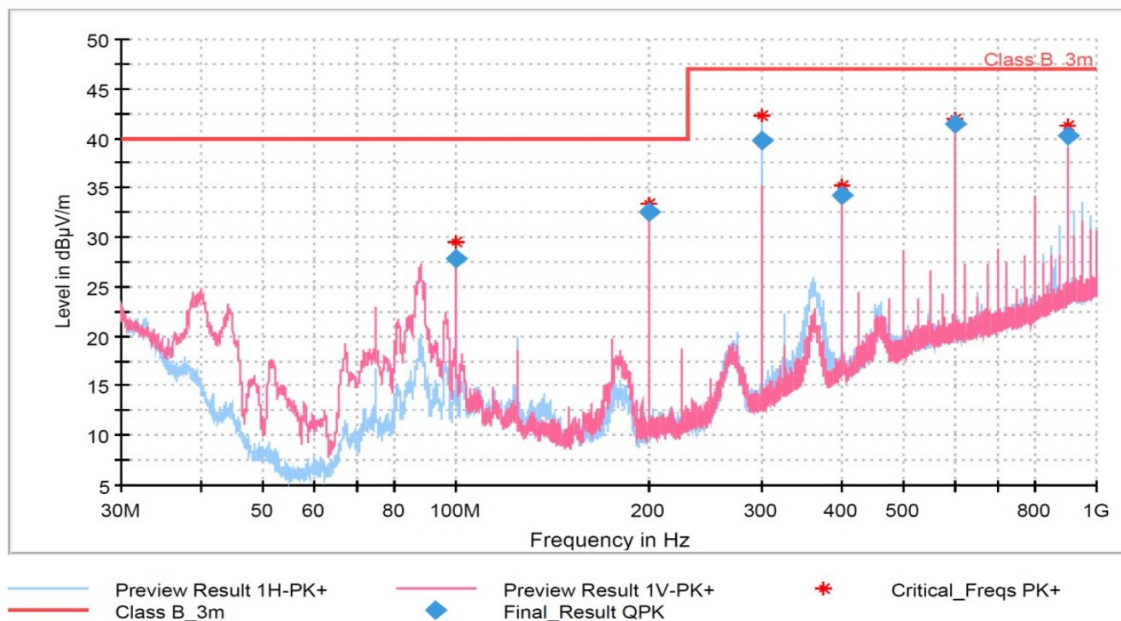


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Figure: Calibrated radiated emissions, 30 MHz - 1 GHz (Reference / ULX4M-LS / EUT vs EN 55032 Class B limit at 3 m).

Reference Device Measurements (SiQ T251-0973/25)

The Intergalaktik chamber transfer function is calibrated against the ULX4M-LS development board (Intergalaktik d.o.o.), formally measured at the accredited SIQ Ljubljana test laboratory (report T251-0973/25). ULX4M-LS is the radiated-emission reference: EUT measurements made in the in-house chamber that sit at or below the ULX4M-LS reference trace at every frequency can be considered compliant by substitution.



Final Result

Frequency (MHz)	QuasiPeak (dBµV/m)	Limit (dBµV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
600.000000	41.36	47.00	5.64	1000.0	120.000	100.0	V	332.0	19.1
900.000000	40.29	47.00	6.71	1000.0	120.000	100.0	H	281.0	22.8
300.000000	39.81	47.00	7.19	1000.0	120.000	100.0	H	0.0	12.4
200.010000	32.47	40.00	7.53	1000.0	120.000	100.0	V	237.0	8.9
99.990000	27.84	40.00	12.16	1000.0	120.000	150.0	V	269.0	10.8
399.990000	34.21	47.00	12.79	1000.0	120.000	150.0	V	2.0	15.1

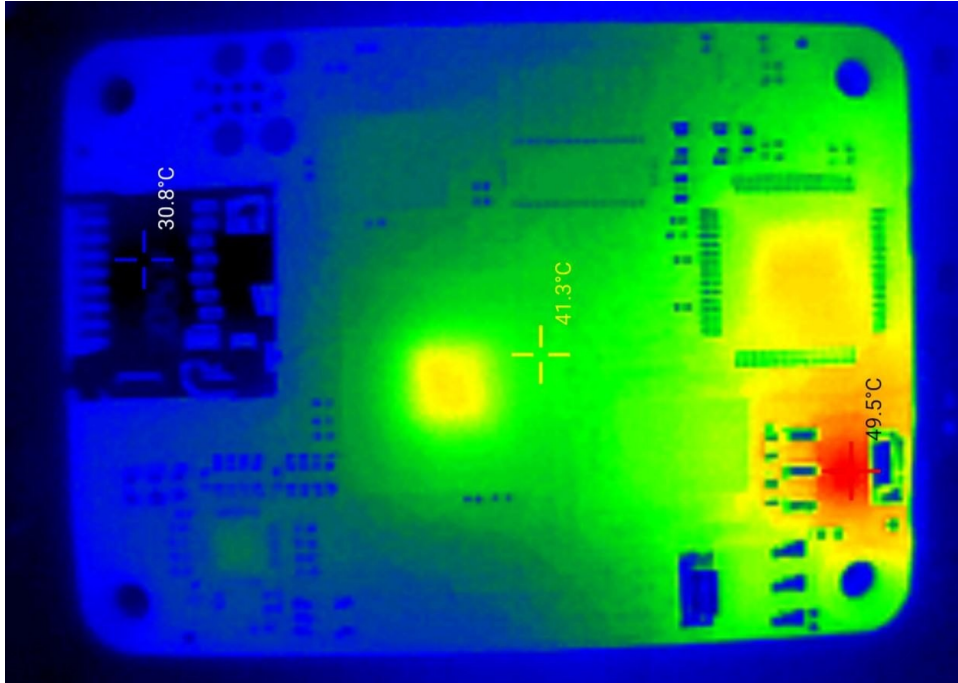
Figure: ULX4M-LS radiated emission spectrum at SIQ Ljubljana (semi-anechoic chamber, 3 m distance). H- and V-polarised preview traces (1H-PK+ / 1V-PK+) and their maximum, against the EN 55032 Class B 3 m Quasi-Peak limit (40 dBµV/m below 230 MHz, 47 dBµV/m above).

Notes & Comments

Thermal camera check

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Thermal camera shows that main source of heat on this board is 3V3 regulator.



LISN Conducted Emissions

Conducted emissions on the AC mains port are measured per EN 55032 / CISPR 32 using a Line Impedance Stabilization Network (LISN) and a TinySA Ultra spectrum analyzer.

Test method

The EUT power input is routed through a CISPR 16-1-2 compliant LISN. It presents a defined $50\ \Omega$ / $50\ \mu\text{H}$ impedance to the device under test on both the L and N conductors of the AC mains over the 150 kHz - 30 MHz band, isolates the EUT from impedance variations of the mains supply, and routes the RF component of each line to a $50\ \Omega$ BNC port for the spectrum analyser. The LISN RF output port is connected (through an optional fixed-value attenuator) to a TinySA Ultra spectrum analyzer, which is swept across the CISPR Band B range 150 kHz - 30 MHz with the parameters listed below.

LISN device

Key specifications:

- **Topology:** two-line CISPR 16-1-2, $50\ \Omega$ / $50\ \mu\text{H}$ per line (L and N), 150 kHz - 30 MHz characterised band.
- **Design:** EEZ LISN r1B2, Intergalaktik-built with reflow-soldered SMT components.
- **EUT capacity:** 10 A continuous at 250 VAC; Schürter PF0030 fused inlet; integrated front-panel toggle switch (M2012S3S2W40) and rocker AC switch (DF62J12S2).
- **RF take-off:** $2 \times 50\ \Omega$ BNC (one per line) for selective differential-mode and common-mode measurement.
- **Transient protection:** TI SG90 gas-discharge surge arrester + 1.5KE400CA bidirectional TVS clamp on the EUT-side rails.
- **Front-panel terminals:** 4 mm safety binding posts on the EUT output for accessory probing; PE binding post bonded to chassis.

Sweep parameters

Span: 0.15 MHz - 30 MHz | RBW: 9 kHz (CISPR Band B) | Detector: peak with 5x peak-hold averaging | Segment step: 1.0 MHz (450 points, 0.1 MHz overlap) | External attenuator (RF output → TinySA Ultra): 0.0 dB | LISN insertion loss (subtracted in post-processing): 0.0 dB.

Acquired segments are stitched into a single frequency-ordered CSV per measurement. The application then converts raw tinySA dBm into calibrated dB μ V at the LISN EUT port using the formula $\text{dB}\mu\text{V} = \text{raw_dBm} + 107 + \text{atten_dB} + \text{LISN_IL_dB}$. CISPR 32 QP and AVG limit lines are overlaid on the resulting trace.

LISN test bench

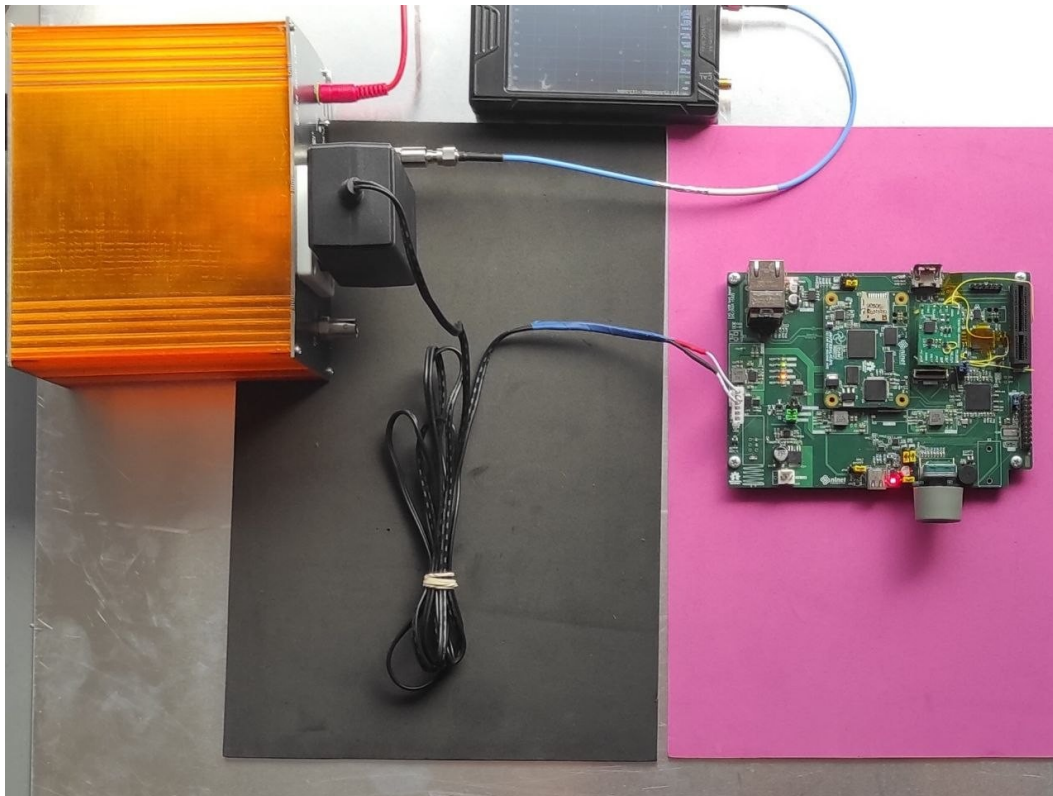


Figure: LISN test bench — EUT power input routed through the LISN, RF output to TinySA Ultra.

Results

The chart below superimposes the EUT trace against the CISPR 32 limits, alongside the empty-mains baseline and (when available) a CE-certified reference device for sanity-check.

Traces included: Envov_CM5H7_PSU (ambient), Envov_CM5H7_connected (eut).

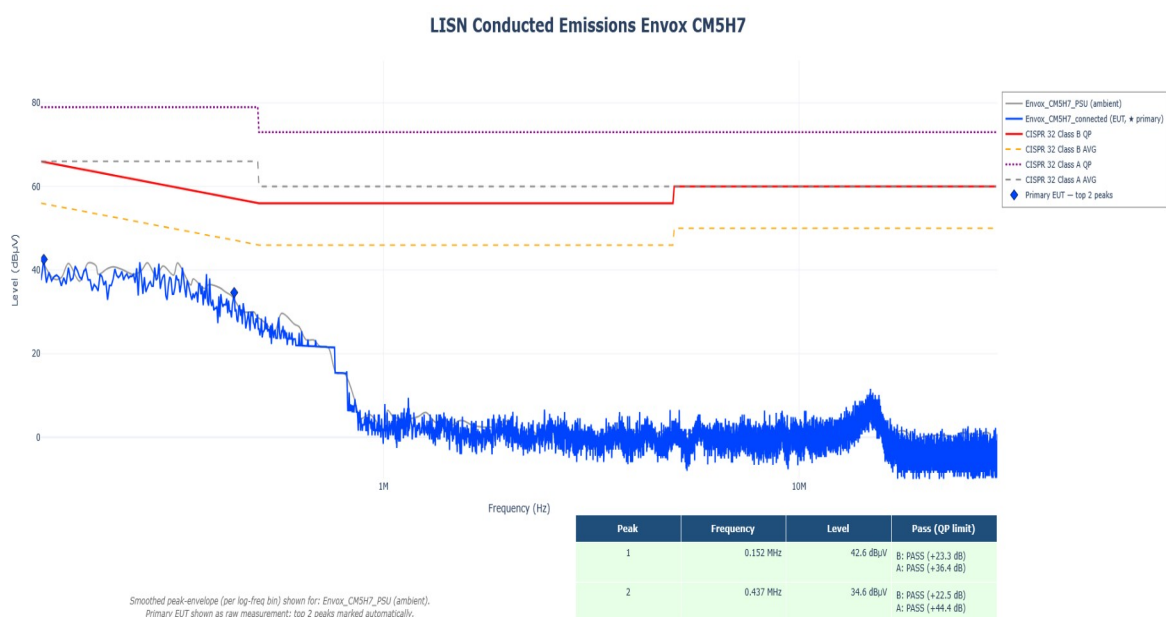


Figure: LISN conducted emissions vs CISPR 32 limits, 150 kHz - 30 MHz.

Peak summary

Top peaks identified on the primary EUT trace (Envox_CM5H7_connected) and their margin against the CISPR 32 quasi-peak limits. Positive margin = below the limit.

Frequency (MHz)	Level (dBμV)	Class B QP (dBμV)	Margin B (dB)	Class A QP (dBμV)	Margin A (dB)	Class B verdict
0.152	42.6	65.9	+23.3	79.0	+36.4	PASS
0.328	40.6	59.5	+18.9	79.0	+38.4	PASS
0.573	26.6	56.0	+29.4	73.0	+46.4	PASS

Verdict: Envox_CM5H7_connected | Class B QP PASS (+18.9 dB @ 0.33 MHz) | Class A QP PASS (+36.4 dB @ 0.15 MHz) | EUT-amb min -10.5 dB (LOW)

Note (Envox_CM5H7_connected): at its closest point the EUT trace sits -10.5 dB relative to the ambient baseline; CISPR asks the EUT to be at least 6 dB ABOVE ambient for the reading to be attributable to the EUT. The margin reported here is therefore limited by the measurement noise floor rather than by the device.

Standards Reference

The following international and European standards form the reference basis for the measurements documented in this report:

Standard	Title	Scope
EN 55032:2015+A1:2020	Electromagnetic compatibility of multimedia equipment	Emission requirements
CISPR 32:2015+A1:2019	Electromagnetic compatibility of multimedia equipment	Emission requirements (international)
EN 55035:2017+A11:2020	Electromagnetic compatibility of multimedia equipment	Immunity requirements
ANSI C63.4	Methods of measurement for radio-noise emissions	Test method reference

Compliance Summary

Based on the emission measurements performed by Intergalaktik and summarised below:

Standard	Test / Scope	Result
EN 55032 / CISPR 32	Conducted emissions, AC mains via LISN — Class B (residential)	PASS (+18.9 dB worst QP margin @ 0.33 MHz)
EN 55032 / CISPR 32	Conducted emissions, AC mains via LISN — Class A (industrial)	PASS (+36.4 dB worst QP margin @ 0.15 MHz)
EN 55032 / CISPR 32	Radiated emissions, 30 MHz – 1 GHz — Class B (indicative)	INDICATIVE (see Combined Calibrated View)

Overall result (Intergalaktik measurements): device is COMPLIANT with the measured conducted-emission limits. Note: the EUT trace does not rise the CISPR-required 6 dB above the ambient baseline, so this margin is noise-floor limited.

Disclaimer

This report documents pre-certification measurements performed in an in-house, non-accredited test chamber by Intergalaktik d.o.o. using the substitution method described in the Methodology section. The values reported here are RELATIVE to a certified reference device (ULX4M-LS, SiQ T251-0973/25) and should be interpreted as engineering estimates.

These results do NOT constitute legal compliance evidence and do NOT replace accredited laboratory testing. For formal certification, please consult a notified accredited test body.